

Simple Reflex Agent

Vacuum Cleaner World — Python Program with Examples

Full Program

```
class SimpleReflexVacuumAgent:
    """
    Simple Reflex Agent for the Vacuum Cleaner World.

    Environment: Two locations, A and B.
    Percept: (location, status) where status is 'Dirty' or 'Clean'.
    Action: 'Suck', 'MoveLeft', 'MoveRight', 'NoOp'
    """

    def __init__(self):
        self.location_A, self.location_B = 'A', 'B'

    def reflex_agent_action(self, location, status):
        """
        Condition-Action Rule (simple reflex -- no memory used):
        - If current location is Dirty -> Suck
        - If in A and Clean -> Move Right
        - If in B and Clean -> Move Left
        """
        if status == 'Dirty':
            return 'Suck'
        elif location == self.location_A:
            return 'MoveRight'
        elif location == self.location_B:
            return 'MoveLeft'
        else:
            return 'NoOp'

def run_example(agent, location, status):
    action = agent.reflex_agent_action(location, status)
    print(f"Percept: (Location: {location}, Status: {status}) --> Action: {action}")
    return action

# ----- MAIN PROGRAM -----
if __name__ == "__main__":
    agent = SimpleReflexVacuumAgent()

    print("=== Simple Reflex Vacuum Cleaner Agent ===\n")

    print("Example 1:")
    run_example(agent, 'A', 'Dirty')      # Expected: Suck

    print("\nExample 2:")
    run_example(agent, 'A', 'Clean')      # Expected: MoveRight

    print("\nExample 3:")
    run_example(agent, 'B', 'Dirty')      # Expected: Suck

    print("\nExample 4:")
    run_example(agent, 'B', 'Clean')      # Expected: MoveLeft
    print("\n=== Simulating a Full Run (5 steps) ===")
```

```

import random

location = 'A'
status_map = {'A': random.choice(['Clean', 'Dirty']),
              'B': random.choice(['Clean', 'Dirty'])}

print(f"Initial Environment: {status_map}\n")

for step in range(1, 6):
    status = status_map[location]
    action = agent.reflex_agent_action(location, status)
    print(f"Step {step}: Location={location}, Status={status}, Action={action}")

    if action == 'Suck':
        status_map[location] = 'Clean'
    elif action == 'MoveRight':
        location = 'B'
    elif action == 'MoveLeft':
        location = 'A'

```

Sample Output

=== Simple Reflex Vacuum Cleaner Agent ===

Example 1:

Percept: (Location: A, Status: Dirty) --> Action: Suck

Example 2:

Percept: (Location: A, Status: Clean) --> Action: MoveRight

Example 3:

Percept: (Location: B, Status: Dirty) --> Action: Suck

Example 4:

Percept: (Location: B, Status: Clean) --> Action: MoveLeft

=== Simulating a Full Run (5 steps) ===

Initial Environment: {'A': 'Dirty', 'B': 'Clean'}

Step 1: Location=A, Status=Dirty, Action=Suck

Step 2: Location=A, Status=Clean, Action=MoveRight

Step 3: Location=B, Status=Clean, Action=MoveLeft

Step 4: Location=A, Status=Clean, Action=MoveRight

Step 5: Location=B, Status=Clean, Action=MoveLeft

This program has been verified to run without errors in Python 3.